



Automated Endotracheal Tube Placement Check Using Semantically Embedded Deep Neural Networks

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Rationale and Objectives: To develop artificial intelligence (AI) system that assists in checking endotracheal tube (ETT) placement on chest X-rays (CXR) and evaluate whether it can move into clinical validation as a quality improvement tool.

Materials and Methods: A retrospective data set including 2000 de-identified images from intensive care unit patients was split into 1488 for training and 512 for testing. AI was developed to automatically identify the ETT, trachea, and carina using semantically embedded neural networks that combine a declarative knowledge base with deep neural networks. To check the ETT tip placement, a “safe zone” was computed as the region inside the trachea and 3–7 cm above the carina. Two AI outputs were evaluated: (1) ETT overlay, (2) ETT misplacement alert messages. Clinically relevant performance metrics were compared against prespecified thresholds of >85% overlay accuracy and positive predictive value (PPV) > 30% and negative predictive value NPV > 95% for alerts to move into clinical validation.

Results: An ETT was present in 285 of 512 test cases. The AI detected 95% (271/285) of ETTs, 233 (86%) of these with accurate tip localization. The system (correctly) did not generate an ETT overlay in 221/227 CXRs where the tube was absent for an overall overlay accuracy of 89% (454/512). The alert messages indicating that either the ETT was misplaced or not detected had a PPV of 83% (265/320) and NPV of 98% (188/192).

Conclusion: The chest X-ray AI met prespecified performance thresholds to move into clinical validation.

Key Words: chest X-ray; endotracheal tube tip position; artificial intelligence.

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INTRODUCTION

Chest radiographs (CXR) are used in the intensive care unit (ICU) to examine and monitor critically ill patients on life-supporting devices. Endotracheal tubes (ETTs) are used to maintain airway patency and lung ventilation. The American College of Radiology recommends that a CXR is performed following intubation to check tube placement (1), with repositioning being required in about 15% of patients (2–6). The desired ETT tip position is within the mid trachea, approximately 5 ± 2 cm above the carina (7), with the head of the patient in a neutral position, and at least 2 cm above the carina (8). If the ETT is too high,

there is a risk of inefficient ventilation or inadvertent extubation and may even cause vocal cord injury (9). On the contrary, selective bronchial intubation may occur if the ETT tip is too low (e.g., close to or lower than carina), causing severe complications, such as partial or complete collapse of the nonventilated lung, hyperinflation of the ipsilateral lung, pneumothorax and even death (2). Based on this literature, a goal of our system will be to issue alerts to physicians if the ETT tip is < 2 cm or > 8 cm above the carina with high sensitivity, i.e., if the ETT tip is outside of the desired range by 1 cm or more.

Tertiary ICUs can generate hundreds of CXRs each day to check for tube placement. Given the high volume of cases and urgent need for intervention if the ETT is misplaced, it is not always practical to wait for radiology reads so ICU physicians may take a preliminary look at the CXR and adjust for the misplaced tube immediately. Due to the low conspicuity of tubes, superposition of anatomy and medical devices, and image quality issues, assessment of tube placement can be challenging without high quality monitors and honed radiologic interpretation skills.

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We propose an artificial intelligence (AI) system that provides two forms of decision support: ETT detection assistance and position check alerts. Our evaluation will focus on supporting the ICU physician at the point of care. The AI system may also benefit radiologists to expedite their reads and improve accuracy.

Our goal is to perform analytic validation of the AI and determine whether it can move into clinical evaluation as a quality improvement tool. Clinically relevant performance metrics will be compared against pre-specified thresholds.

MATERIALS AND METHODS

Image Data Set

This retrospective study was approved by our institutional review board (IRB #11-000126). Our study cohort included consecutive patients from medical and surgical ICUs across our hospital system between April 2018 and September 2019. CXRs were bedside anteroposterior images from adult patients (aged ≥ 18 years) with or without ETT. For each patient, CXR images from up to separate three examinations were included. Data were de-identified to comply with Health Insurance Portability and Accountability Act and IRB requirements.

Using an in-house developed imaging research platform (10), CXR images were manually annotated by trained image analysts, with supervision by three cardiothoracic radiologists who have been in practice for 10 to over 25 years. Following training by the radiologists, image analysts reviewed cases with radiologists if uncertain about the location of the ETT or anatomic landmarks. The trachea was outlined, the carina point marked, and the ETT traced.

Two thousand CXR images were annotated and split into 1488 for training and 512 for testing (approx. 3:1). Training and test sets had similar proportions of cases with and without ETT, approximately 60:40.

Algorithm Development

The AI automatically identifies the ETT, trachea, and carina using a “semantically embedded neural network” that combines a declarative knowledge base with deep convolutional networks (DCNNs). Each node of the semantic network represents an entity (anatomy or ETT) and defines a segmentation method (DCNN) and expected size and shape features. The connections of the semantic network reflect expected spatial relationships between the entities. Nodes were created for the ETT, ETT tip, trachea, carina, and safe zone. Based on the relationships defined in the semantic network, segmentation proceeds in a hierarchical fashion, with the segmentation of the trachea spatially guiding the CNNs detecting the carina and ETT. Spatial relationships are used directly to define the ETT tip and safe zone regions from

these segmentation results. As described in (11), the segmentation runs for each node to generate candidate image regions. The candidate that best satisfies spatial, size, and shape constraints is then matched to that node, excluding extraneous fragments generated by the DCNN. The spatial constraints are also used to define a search area for the node, based on previously identified nodes (e.g., the detected trachea guides image cropping for DCNN segmentation of the ETT).

Automatic trachea segmentation is performed using U-Net DCNN with five encoder and five decoder blocks. CXRs were rescaled to 512×512 pixels and intensity normalized. The DCNN was trained using Adam optimization with Dice coefficient loss and a learning rate of 0.00005. Two DCNNs were trained for ETT segmentation, one as for the trachea and another using a cropped image around the trachea with contrast limited adaptive histogram equalization and their outputs were combined.

Carina point localization is performed using a regression predictive model based on the VGGNet architecture with five blocks and 16 weight layers. The DCNN output is a 2×1 array of points representing the coordinates of the carina.

The “safe zone” is the region where the ETT tip should be located if correctly placed. It is defined using spatial relationships that describe a region inside the trachea and between 3 and 7 cm above the carina. This range is based on desired position of an ETT tip 5 ± 2 cm above the carina in a neutral neck position (7). The system will issue an alert if the ETT tip is outside of this safe zone, so the AI is designed to be highly (overly) sensitive to misplacement of tubes, given that our goal is to issue alerts for ETT tips <2 cm or >8 cm above the carina.

AI System Outputs

The AI generated CXR overlays showing the ETT path and measurement in cm from the ETT tip to the carina. It also displayed one of three possible ETT messages: (1) “Found” (ETT tip was determined to be in the safe zone), (2) “Position Alert” (ETT tip was not in the safe zone or the AI could not determine the safe zone), (3) “Not Found.”

Messages 2 and 3 are intended to prompt a check by the radiologist or physician.

Evaluation

The two forms of AI decision support were evaluated: (1) the ETT overlay, (2) the alert messages. Evaluations were performed over the entire set of test cases ($N = 512$) and the subgroup with ETT present ($N = 285$). In the ICU, where alerts will be presented to the physician at the point of care, we expect a high proportion of cases will have ETTs present, therefore, we prespecified evaluation metric targets for the ETT subgroup to confirm the system is ready to move into clinical evaluation.

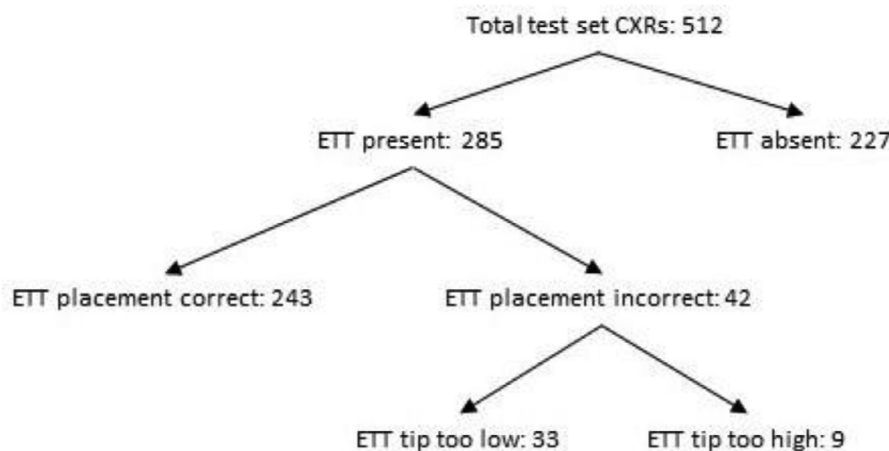


Figure 1. CXR test set composition. CXR, chest X-ray.

The ETT overlay was evaluated in terms of percentage of cases where the overlay was accurate, i.e., the ETT was detected and the tip coordinates were within 10 mm of reference (if the ETT was present), or no ETT was detected (if the ETT was not present). The prespecified performance requirement on this metric was $>85\%$ accurate to move into clinical evaluation.

For alert messages, we define a positive output (alert) as either a “Position Alert” message (misplaced ETT) or a “Not Found” message (missing ETT), and a negative output (no alert) as a “Found” message (ETT present and correctly placed). When the AI output is positive, a true positive (TP) requires (ETT tip outside trachea) or (tip < 2 cm above carina) or (tip > 8 cm above carina) or (ETT is missing), otherwise it is a false positive (FP). When the output is negative, a false negative (FN) requires (ETT tip point outside trachea) or (tip < 2 cm above carina) or (tip > 8 cm above carina), otherwise it is a true negative (TN). Positive predictive value ($PPV = TP / [TP + FP]$) and negative predictive value ($NPV = TN / (TN + FN)$) metrics were computed to give a sense of trustworthiness from a physician perspective. Pre-specified thresholds for the ETT subgroup were $PPV > 30\%$ and $NPV > 95\%$, respectively, to move into clinical evaluation. The PPV threshold considers the deliberate design of the system to be overly sensitive to misplaced tubes and is based on the premise that an ICU physician will tolerate 1 to 2 FPs for each TP alert. To be safe to move into clinical evaluation, the NPV must be high, i.e., avoid missed alerts when tubes are misplaced.

We also performed a sub-analysis of sensitivity to misplaced tubes. The primary analysis included both misplaced and absent ETTs as positives. Misplaced ETTs were identified in the test set based on the reference annotations as follows: (ETT tip outside trachea) or (tip < 2 cm above carina) or (tip > 8 cm above carina). For ETT detected, the accuracy of tip location determined by the AI was computed as the mean absolute difference and standard deviation from the reference annotated manually.

RESULTS

The composition of the test set is shown in [Figure 1](#). It provides a breakdown of cases with and without ETT and whether the ETT was correctly or incorrectly positioned, based on the reference annotations according to the definitions above.

Overlay (detection assist) accuracy was 89% in the test set, i.e., in $(233 + 221)/512$ test cases. Two hundred seventy-one of 285 tubes were detected in the image, having a mean absolute difference of 0.72 ± 1.05 cm in tip position from the reference annotation, and 233 of these had the tip within 10 mm of the reference, i.e., 233 overlays were accurate when the ETT was present. The system (correctly) did not generate an ETT overlay in 221/227 CXRs where the tube was absent.

The system PPVs/NPVs are shown in [Table 1](#). The pre-specified PPV and NPV thresholds for the ETT subgroup were met. False positive was defined as correctly placed ETT classified by the AI as misplaced ETT or missing ETT (i.e., not detected). Of the 55 FPs, 14 were due to missed ETT detections and 41 due to unnecessary alerts for misplacement according to the reference.

The system sensitivity to misplaced tubes was 95% . There were 42 misplaced ETTs in the test set, of which 40 were positive alerts by the system. The two FNs were due to incorrect localization of the ETT tip and/or carina.

Three examples are shown in [Figures 2–4](#), including the original CXR and the AI output image automatically generated by the AI. The output image shows the ETT overlay

TABLE 1. Positive Predictive Value (PPV) and Negative Predictive Value (NPV) of Chest X-Ray AI Alerts

	All Test Cases (N = 512)	ETT Present Subgroup (N = 285)
PPV	83% (TP = 265, FP = 55)	42% (TP = 40, FP = 55)
NPV	98% (TN = 188, FN = 4)	99% (TN = 188, FN = 2)

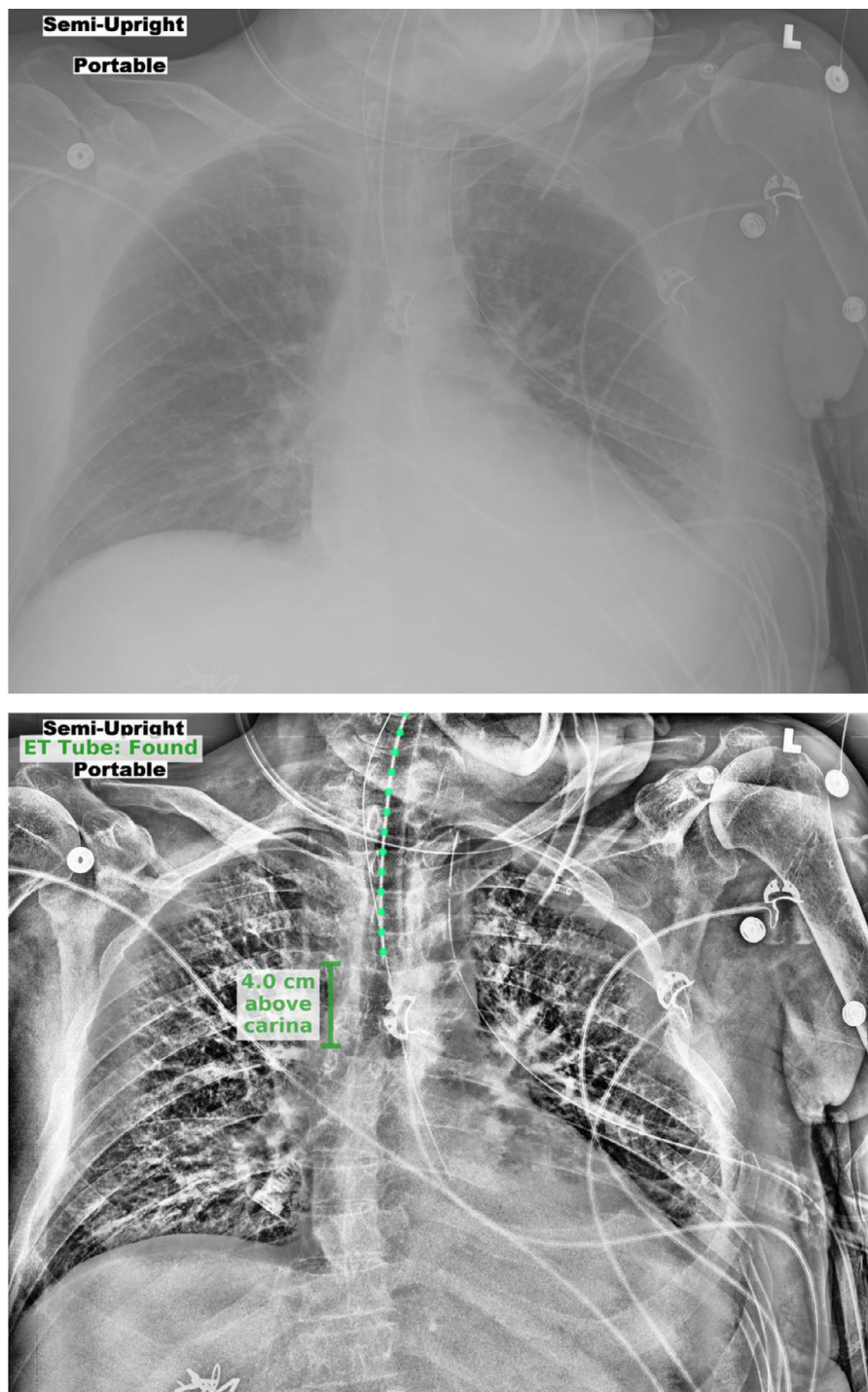


Figure 2. Original CXR and the AI output image for a correctly placed ETT. AI, artificial intelligence; CXR, chest X-ray; ETT, endotracheal tube.

and alert message on the contrast enhanced image generated by the system for input to the DCNN. The example images show an ETT detected by the system with correct placement (Fig 2), and two with ETT alerts, one with the ETT tip too low relative to the carina (Fig 3) and one too high (Fig 4). Figure 5 shows a false positive case flagged by the AI, which incorrectly identified the ETT tip at the point where the projection of NG tube crossed it, resulting in an unnecessary alert

message that the ETT tip was too high, when in fact the ETT extended downward to a safe distance from the carina.

DISCUSSION

Deep learning (DL) has been widely used in image processing since the successful application of CNNs to image classification and object recognition (12). Various DL approaches

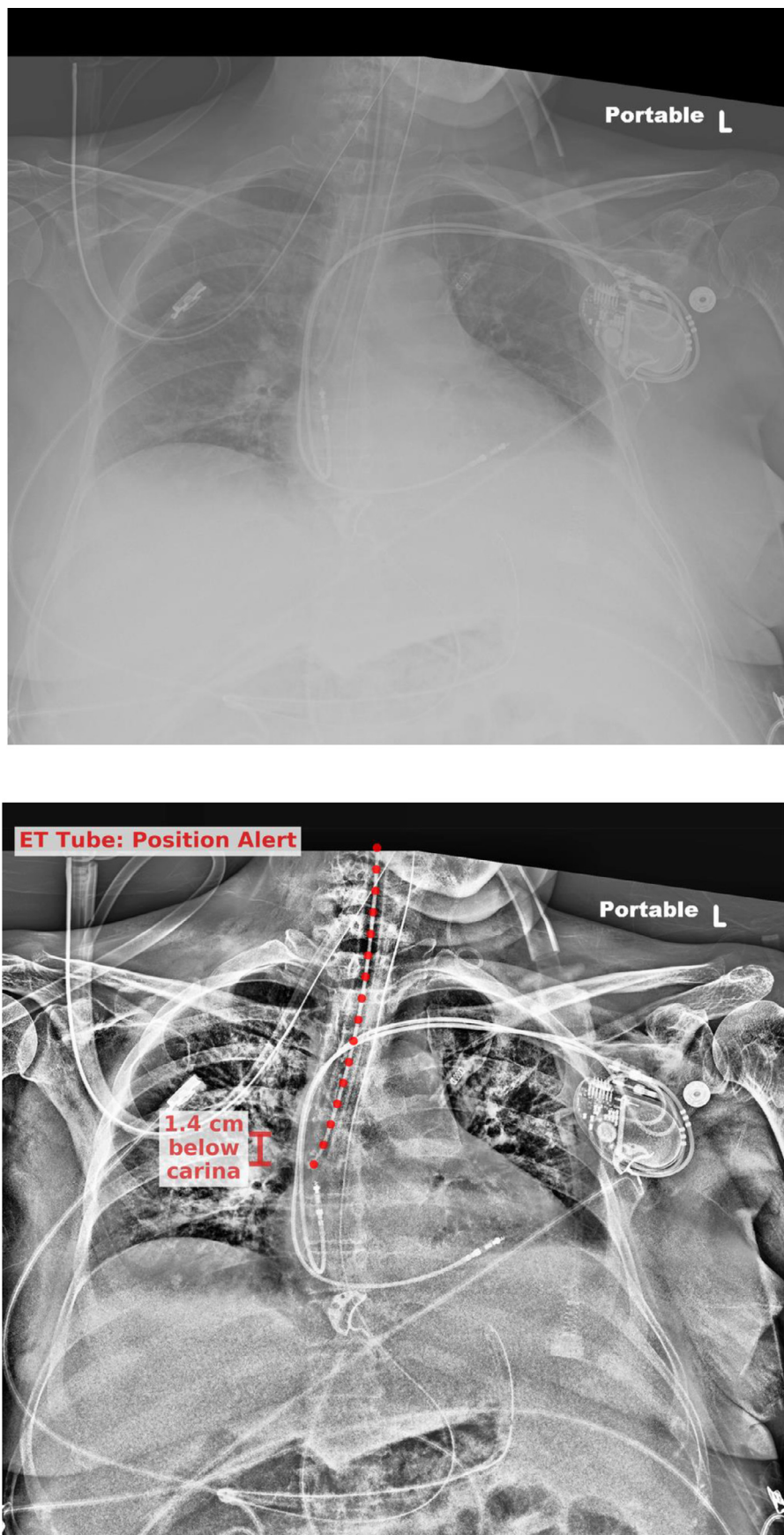


Figure 3. Original CXR and the AI output image for an ETT with tip placed too low. AI, artificial intelligence; CXR, chest X-ray; ETT, endotracheal tube

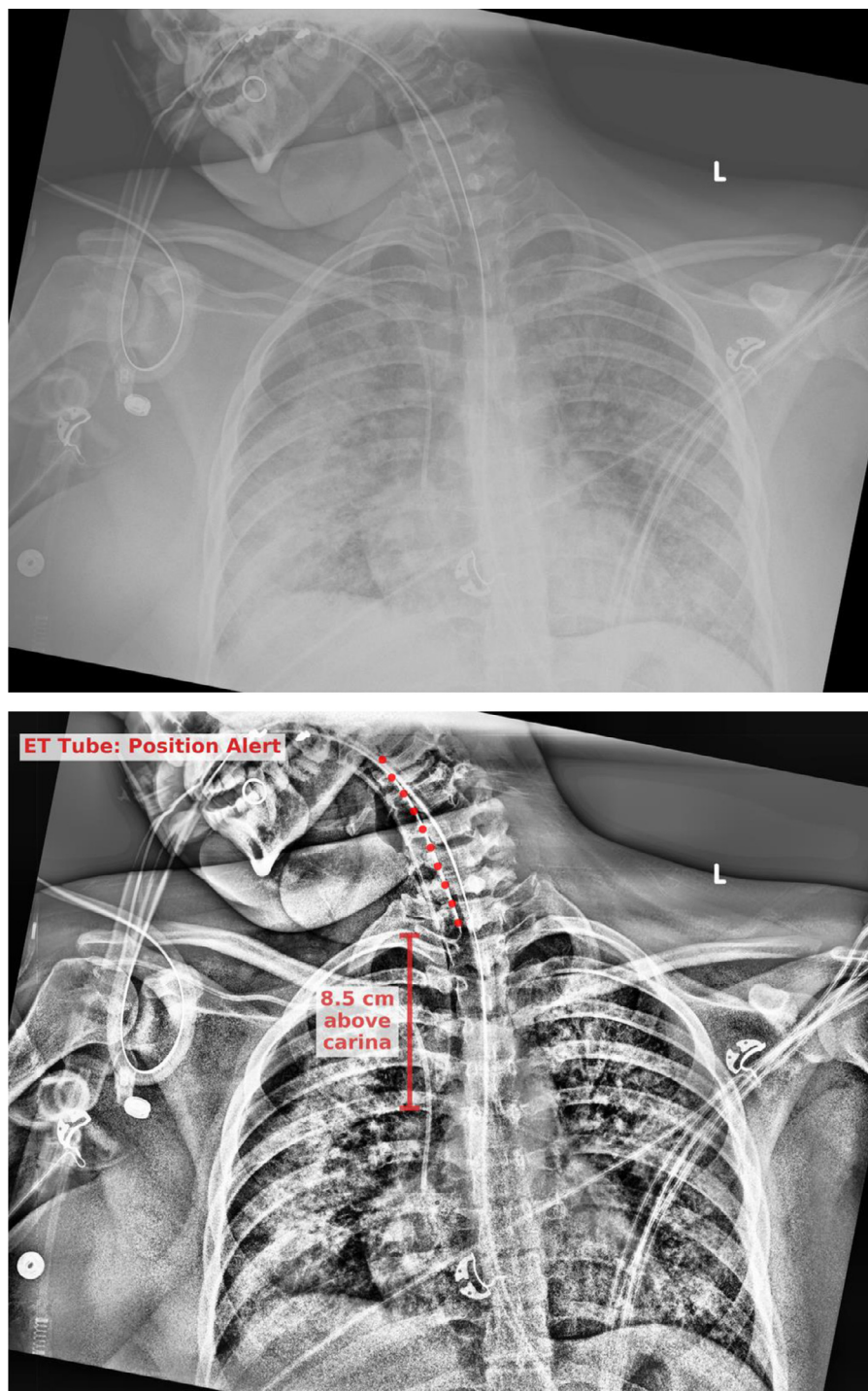


Figure 4. Original CXR and the AI output image for an ETT with tip placed too high. AI, artificial intelligence; CXR, chest X-ray; ETT, endotracheal tube.

have been applied to many different aspects of CXR, ranging from image quality control (13) and detection of the orientation of CXR images (14,15) to more complicated applications, such as anatomic segmentation (16,17), disease classification (18–22), and ETT detection (23–27). Several reports have been published for automated detection of the ETT on CXR. Computer-aided detection

methods have yielded a true-positive rate of 0.93 using template matching and region growing segmentation (28) and an area under the receiver operating characteristic curve (AUC) of 0.94 using image enhancement and non-linear spatial filtering (29). Using a DL approach with a U-Net-like architecture, Frid-dar *et al.* (30) achieved an AUC of 0.99 in classifying CXRs having an ETT or not.

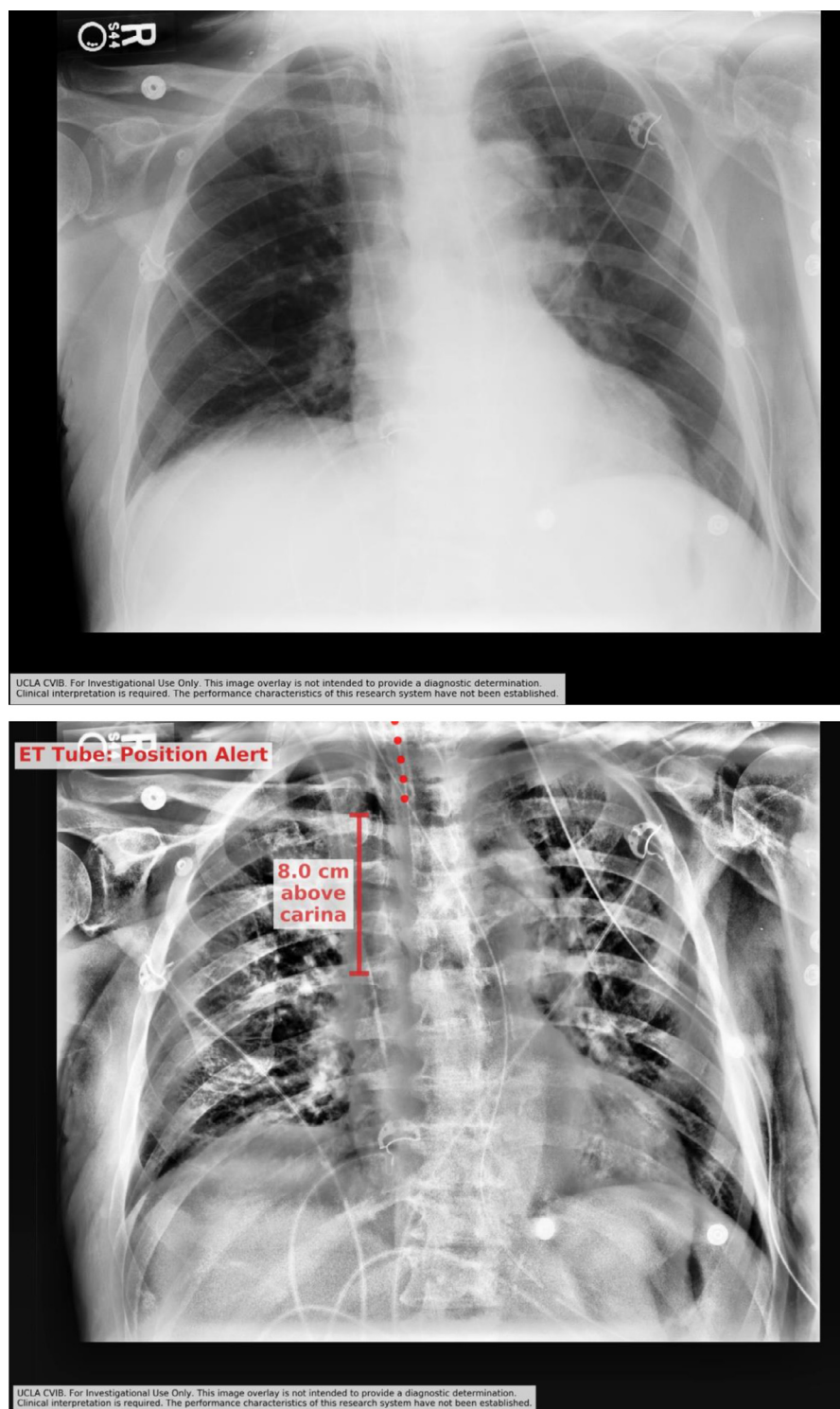


Figure 5. Original CXR and the AI output image for a false-positive case in which the ETT tip position was flagged by the AI as being too high even though the ETT was placed properly. AI, artificial intelligence; CXR, chest X-ray; ETT, endotracheal tube.

They also proposed an approach for synthesizing ET tubes to generate ground-truth data for ETT detection and segmentation. These systems focused on ETT detection only, without placement checking.

Several studies have included assessment the ETT tip position relative to the carina in their evaluation. Using a GoogLeNet CNN and 300 labeled images, Lakhani (23) achieved an AUC of 0.94 on classifying presence vs absence of the

ETT, but a more modest accuracy of 0.81 in differentiating low versus normal positioning of the ETT. Lakhani *et al.* (24) evaluated an ETT-carina distance estimation using the Inception V3 deep CNN architecture and demonstrated excellent interrater agreement (intraclass correlation coefficient >0.8) between AI and radiologists. The sensitivity was 90.1% and specificity 92.4% in detecting ETT tip located in <2 cm above the carina. However, the sensitivity was only 66.5% and specificity 99.2% when assessing ETT-carina distance ≥ 7 cm. Using the open-source MIMIC Chest X-Ray dataset and 3 CNN algorithms, Kara *et al.* (25) reported the accuracy, sensitivity, specificity, PPV, NPV, and AUC of 97.14%, 97.37%, 96.89%, 97.12%, 97.15%, and 99.58% respectively, upon a five-fold cross-validation to classify the presence or absence of the ETT, and the model-predicted locations of the carina and the distal tip of ETT were estimated having a median error of 0.46 cm and 0.60 cm from manual ground-truth annotations, respectively. Harris *et al.* (26) trained a bounding box-based CNN for localization of the ET tube and the carina, with a mean difference of 0.7 cm between the model-predicted ETT-carina distance and the ground-truth measurement agreed by two experienced radiologists in 200 CXR images.

Our study was not designed to compare against other systems, but the metrics are favorable to the prior literature in both classification problems, presence/absence of ETT and assessment of ETT tip position, and in terms of the number of CXRs used for testing. Performing a complete segmentation of the ETT using fully annotated data gives our system the additional capability of generating an overlay to assist the physician with their visual interpretation, and allows explicit (explainable and adjustable) rules for checking the ETT tip position, as opposed to a black box approach of classifying ETTs as correctly or incorrectly placed.

The combination of neural and semantic networks in our system was beneficial in this application. The deep neural networks allow learning of complex or subtle patterns (e.g., relating to the gradient profiles and trajectories of the ETT), that are difficult to describe as explicit features, to be learned directly from the data. The semantic networks are complementary in allowing direct expression of placement check rules. This is advantageous in this setting because there are relatively few misplaced ETT examples to train on for a data driven approach. There is also the benefit of explainability of the decision when a tube is declared as misplaced which aids in user confidence and adoption. The semantic network also allowed constraints, such as continuity and length of the ETT to be applied and contributed to the reliability of the detection and segmentation.

The current implementation of the AI system is that alert messages are provided for cases with ETT misplacement and ETT missing, which is appropriate for our initial clinical application to cases that are ordered for checking ETT placement in the ICU, and as such our data analysis was focused on the subset with ETT present. If the system were deployed

for all chest X-rays, where a greater proportion would have no ETT, then issuing alerts for ETT missing may not be appropriate and the system analysis and operating points revised accordingly.

Our prespecified metric thresholds were defined to allow the AI to move into a clinical evaluation phase. With ongoing system refinement, this overlay accuracy metric should eventually be above 90%, but >85% was considered sufficient at this stage (89% was achieved). The prespecified PPV target was lower than the NPV because it is crucial to alert on misplaced ETTs and we are willing to accept false positives. The system is being designed to detect misplaced ETTs outside of the range 2–8 cm above the carina, but internally the system issues an alert if the ETT tip is outside of a 3–7 cm range, i.e., we are designing it to be overly sensitive and as such we were willing to accept a PPV above 30% to move into clinical evaluation (42% was achieved). While false positives can be inefficient and over time cause alert fatigue, they do not pose a direct risk to patient management. We will ultimately strive for a low false positive rate, but this target was considered sufficient to permit moving into the initial clinical evaluation where physician acceptance can be further studied. Our prespecified NPV threshold was >95% because the biggest safety concern is a misplaced ETT, i.e., there is a very low tolerance for false negatives. We achieved an NPV of 99% in this small data set of 285 ETT cases with 42 misplaced tubes and the NPV will need to be maintained very close to 100% in larger studies before being cleared for clinical use. It should be noted that the 15% (42/285) of cases with misplaced ETT by reference in our data set is consistent with the incidences of misplaced tubes reported in the literature in clinical practice.

There are some limitations to this study. First, training and test samples were obtained from ICU facilities of a single institution where acquisition protocols and equipment are usually well standardized. The performance of AI might be affected if the trained models were applied to CXR images obtained from other institutions. Another potential limitation is the heterogeneity of samples as they were collected from both medical and surgical ICUs. In comparison to typical medical ICU patients, surgical ICU patients will generally have more life-supporting lines and tubes that could represent a different level of complexity and challenge to most AI models and algorithms. Our prespecified decision support metric benchmarks were met, but this is not a clinical validation of our system. A simple analysis was performed to make a practical decision about moving into the next phase of clinical evaluation. We believe it is important for radiology AI evaluations to have prespecified targets for clinically relevant metrics if we are to advance such technology into clinical practice. The intended use in our next phase of clinical evaluation is for healthcare quality improvement providing ETT detection assist overlays and position check alerts. The system outputs will be made available to both ICU physicians at the point of care and reporting radiologists and the impact in each setting will be evaluated. The physician remains responsible for the final

diagnosis and patient management decisions. Our conclusion for this phase of our development is that the AI decision support performance is sufficient to move into clinical implementation and further evaluation.

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